

DAY-2 NUMPY

```
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[1]: a=[1,2,3,4,5]
     print(a)
     [1, 2, 3, 4, 5]

[4]: import numpy as np
     arr=np.array([1,2,3,4,5])
     print(arr)
     [1 2 3 4 5]

[9]: a=[1,2,3]
     b=[4,5,6]
     c=[]
     for i in range(len(a)):
         c.append(a[i]+b[i])
     print(c)
     [5, 7, 9]

[11]: import numpy as np
     a=np.array([1,2,3])
     b=np.array([4,5,6])
     print(a+b)
     print(a-b)
     print(a*b)
     print(a/b)
     print(a%b)
     print(a**b)
```

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[5 7 9]
[-3 -3 -3]
[ 4 10 18]
[0.25 0.4 0.5 ]
[1 2 3]
[ 1 32 729]

[15]: # numpy ndimensional array

1 dimensional array

2.2dimensional array

3. multidimensional array

all the three are used here
all the nd attributes are used
we dont have shapes in multi dimensional array

[32]: import numpy as np
     arr=np.array([1,2,3,4,5])
     arr=np.array([[1,2,3],[4,5,6]])
     arr=np.array([[[1,2],[3,4],[5,6]]])
     print(arr)
     print(ar)
     print(a)
     print(arr.shape)
     print(ar.shape)
     print(a.shape)
     print(arr.ndim)
```

```
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print(a.ndim)
print(arr.size)
print(arr.size)
print(a.size)
print(arr.dtype)
print(a.dtype)
print(a.dtype)

[1 2 3 4 5]
[[1 2 3]
 [4 5 6]]
[[[1 2]
  [3 4]
  [5 6]]]
(5,)
(2, 3)
(1, 3, 2)
1
2
3
5
6
6
int64
int64
int64

zeros and ones fun in 1,2 and 3 rd array

[41]: import numpy as np
      arrp.array([1,2,3,4,5])
      b=np.zeros(5)
      print(a)
      print(b)
      c=np.zeros((2,2))
```

```
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[41]: import numpy as np
      arrp.array([1,2,3,4,5])
      b=np.zeros(5)
      print(a)
      print(b)
      c=np.zeros((2,2))
      d=np.ones(3)
      print(d)
      e=np.ones((2,3))
      print(e)
      print(np.arange(1,10,2))
      print(np.linspace(0,20,5))
      print(np.random.rand(5))
      print(np.random.randint(0,100,5))
      print(np.random.randn(5))

[1 2 3 4 5]
[0. 0. 0. 0. 0.]
[1. 1. 1.]
[[1. 1. 1.]
 [1. 1. 1.]]
[1 3 5 7 9]
[ 0.  5. 10. 15. 20.]
[0.35121795 0.56048194 0.42788218 0.29598125 0.49039901]
[77 80 24 82 65]
[ 0.43327571  0.08858298 -0.96264243  0.42916562 -0.06740362]

INDEXING AND SLICING

[42]: import numpy as np
      arrp.array([10,20,30,40,50,60,70])
      #1d array indexing
      print(a[2])
      print(a[-1])
      b=np.array([[1,2,3],
                  [4,5,6],
                  [7,8,9]])
      #2d array indexing
      print(b[1][1])
      #1d slicing
      print(a[2:6])
      print(a[:4])
      print(a::1])
      #2d slicing
      print(b[1])
      print(b[:,1])
      print(a[a>50])
      print(a[a<50])
      print(a[0,4,3,6]])
      print(a[a%2==0])
      a[4]=1000
      print(a)

30
70
5
[30 40 50 60]
[10 20 30 40]
```

```
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INDEXING AND SLICING

[42]: import numpy as np
      arrp.array([10,20,30,40,50,60,70])
      #1d array indexing
      print(a[2])
      print(a[-1])
      b=np.array([[1,2,3],
                  [4,5,6],
                  [7,8,9]])
      #2d array indexing
      print(b[1][1])
      #1d slicing
      print(a[2:6])
      print(a[:4])
      print(a::1])
      #2d slicing
      print(b[1])
      print(b[:,1])
      print(a[a>50])
      print(a[a<50])
      print(a[0,4,3,6]])
      print(a[a%2==0])
      a[4]=1000
      print(a)

30
70
5
[30 40 50 60]
[10 20 30 40]
```

```
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30
70
5
[30 40 50 60]
[10 20 30 40]
[10 20 30 40 50 60 70]
[4 5 6]
[2 5 8]
[60 70]
[10 20 30 40]
[10 50 40 70]
[10 20 30 40 50 60 70]
[ 10 20 30 40 1000 60 70]

[60]: # the above is separated and shown below

[43]: import numpy as np
      a=np.array([10,20,30,40,50,60,70])
      #1d array indexing
      print(a[2])
      print(a[-1])

30
70

[44]: import numpy as np
      b=np.array([[1,2,3],
                  [4,5,6],
                  [7,8,9]])
      #2d array indexing
      print(b[1][1])

5
```

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[45]: import numpy as np
      a=np.array([10,20,30,40,50,60,70])
      #1d array indexing
      print(a[2:6])

[30 40 50 60]

[46]: import numpy as np
      a=np.array([10,20,30,40,50,60,70])
      #1d array indexing
      print(a[:4])

[10 20 30 40]

[47]: import numpy as np
      a=np.array([10,20,30,40,50,60,70])
      #1d array indexing
      print(a[::1])

[10 20 30 40 50 60 70]

[48]: import numpy as np
      b=np.array([[1,2,3],
                  [4,5,6],
                  [7,8,9]])
      #2d array indexing
      print(b[1])

[4 5 6]

[49]: import numpy as np
      b=np.array([[1,2,3],
                  [4,5,6],
                  [7,8,9]])
```

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#2d array indexing
print(b[:,1])

[2 5 8]

[50]: import numpy as np
      a=np.array([10,20,30,40,50,60,70])
      #1d array indexing
      print(a[a>50])

[60 70]

[51]: import numpy as np
      a=np.array([10,20,30,40,50,60,70])
      #1d array indexing
      print(a[a<50])

[10 20 30 40]

[52]: import numpy as np
      a=np.array([10,20,30,40,50,60,70])
      #1d array indexing
      print(a[[0,4,3,6]])

[10 50 40 70]

[53]: import numpy as np
      a=np.array([10,20,30,40,50,60,70])
      #1d array indexing
      print(a[a%2==0])

[10 20 30 40 50 60 70]
```

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```
[54]: import numpy as np
      a=np.array([10,20,30,40,50,60,70])
      #1d array indexing
      a[4]=1000
      print(a)

[ 10  20  30  40 1000  60  70]
```

mathematical operation in numpy

3 types of broadcasting is given here

```
[55]: import numpy as np
      #scalar broadcasting
      a=np.array([1,2,3])
      print(a+10)

[11 12 13]
```

```
[56]: import numpy as np
      #two dimensional array broadcasting
      b=np.array([[1,2,3],[4,5,6]])
      print(b+10)

[[11 12 13]
 [14 15 16]]
```

```
[59]: import numpy as np
      #two dimensional array broadcasting (ROW BROADCASTING)
      b=np.array([[1,2,3],[4,5,6]])
      print(b+a)
```

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```
##here a is a 1 dimensional array ...so a is adding in both the rows of b
##dimension should be equal else it will show error like..a has 3 row and b also has 3 rows

[[2 4 6]
 [5 7 9]]
```

matrix operation

```
[61]: import numpy as np
      a=np.array([[1,2],[3,4]])
      b=np.array([[4,3],[2,1]])
      print(a*b)

[[4 6]
 [6 4]]
```

```
[62]: import numpy as np
      a=np.array([[1,2],[3,4]])
      b=np.array([[4,3],[2,1]])
      print(a+b)

[[5 5]
 [5 5]]
```

[]: ## here both dot , matmul and @ symbol give the same ans

```
[63]: import numpy as np
      a=np.array([[1,2],[3,4]])
      b=np.array([[4,3],[2,1]])
      print(np.dot(a,b))

[[ 8  5]
 [20 13]]
```

FCAT- import numpy as np

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```
[64]: import numpy as np
      a=np.array([[1,2],[3,4]])
      b=np.array([[4,3],[2,1]])
      print(a@b)

[[ 8  5]
 [20 13]]
```

```
[66]: import numpy as np
      a=np.array([[1,2],[3,4]])
      b=np.array([[4,3],[2,1]])
      print(np.matmul(a,b))

[[ 8  5]
 [20 13]]
```

[67]: # to transpose an array

```
[68]: import numpy as np
      a=np.array([[1,2],[3,4]])
      b=np.array([[4,3],[2,1]])
      print(a.T)# row becomes column

[[1 3]
 [2 4]]
```

[69]: #universal mathematical function

```
import numpy as np
a=np.array([4,9,16])
print(np.sqrt(a))

[2.  3.  4.]
```

```
[70]: import numpy as np
```

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```
[70]: import numpy as np
      b=np.array([-5,-6,-2,-8])
      print(np.log(a))
      [1.38629436 2.19722458 2.77258872]

[71]: import numpy as np
      b=np.array([-5,-6,-2,-8])
      print(np.exp(a))
      [5.45981500e+01 8.10308393e+03 8.88611052e+06]

[72]: #round function
      arr=np.array([1.234,5.674])
      print(np.round(arr,2))
      [1.23 5.67]

[5]: import numpy as np
      arr=np.array([1.234,5.674])
      print(np.floor(arr))
      [1. 5.]

[6]: import numpy as np
      arr=np.array([1.234,5.674])
      print(np.ceil(arr))
      [2. 6.]

[50]: #statistical function
      ### axis can be used in mean median and so on ....

      import numpy as np
      a=np.array([10,2,3,40,50])
```

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```
      print(np.mean(a))
      21.0

[8]: import numpy as np
      a=np.array([10,2,3,40,50])
      print(np.median(a))
      10.0

[9]: import numpy as np
      a=np.array([10,2,3,40,50])
      print(np.std(a))
      20.039960079800558

[10]: import numpy as np
      a=np.array([10,2,3,40,50])
      print(np.max(a))
      50

[11]: import numpy as np
      a=np.array([10,2,3,40,50])
      print(np.min(a))
      2

[12]: import numpy as np
      a=np.array([10,2,3,40,50])
      print(np.percentile(a,100))
      50.0
```

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```
[13]: import numpy as np
      a=np.array([10,2,3,40,50])
      print(np.percentile(a,26))
      3.2800000000000002

[14]: import numpy as np
      a=np.array([10,2,3,40,50])
      print(np.percentile(a,25))
      3.0

[15]: import numpy as np
      a=np.array([10,2,3,40,50])
      print(np.percentile(a,50))
      10.0

[16]: import numpy as np
      a=np.array([10,2,3,40,50])
      print(np.var(a))
      401.6

[17]: import numpy as np
      a=np.array([10,2,3,40,50])
      print(np.argmax(a))
      #return the min index value
      1

[18]: import numpy as np
```

```
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[18]: import numpy as np
      a=np.array([10,2,3,40,50])
      print(np.argmax(a))
      #return the max index value
      4

[24]: b=np.array([1,2,3,4,5])
      c=np.array([2,4,6,8,10])
      print(np.corrcoef(a,b))
      # if it returns positive result it means positive relationship
      [[1. 0.83272222]
       [0.83272222 1.]]

[22]: b=np.array([1,2,3,4,5])
      c=np.array([10,8,6,4,2])
      print(np.corrcoef(b,c))
      # if it returns positive result it means positive relationship
      [[1. -1.]
       [-1. 1.]]

[25]: #aggregate function
      d=np.array([[1,3,4],[5,6,8]])
      print(np.sum(d))
      27

[26]: d=np.array([[1,3,4],[5,6,8]])
      print(np.sum(d,axis=0))
      #axis=0 sum column wise
      [ 6  9 12]
```

```
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#axis=0 sum column wise
[ 6  9 12]

[28]: d=np.array([[1,3,4],[5,6,8]])
      print(np.sum(d,axis=1))
      #axis=1 sum row wise
      [ 8 19]

[49]: import numpy as np
      b=np.array([[14,25,36],
                  [42,54,85],
                  [71,68,99]])
      print(np.mean(b))
      a=np.sum(b,axis=1)
      print(np.max(a))
      print((b - np.min(b)) / (np.max(b) - np.min(b)))
      54.888888888888886
      238
      [[0. 0.12941176 0.25882353]
       [0.32941176 0.47058824 0.83529412]
       [0.67058824 0.63529412 1.]]

[ ]:
```